PROJECT PROPOSAL

SDF04 – Object-Oriented Programming

Project Title: Smart Gym Workout Recommender and Progress Tracker	
Subject Code	SDF04
Subject Title	Object-Oriented Programming (Python & Database)
Application Type	Desktop / Web GUI Application (Non-Console)
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Course / Year / Section	BSCS / LFCA311M111
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1. INTRODUCTION / BACKGROUND

Starting a structured fitness journey often presents an overwhelming challenge for beginners and independent gym-goers. Without access to costly personal trainers, individuals rely on fragmented manual processes—using a basic calculator to compute their Body Mass Index (BMI), browsing static websites for random exercise lists, using a phone timer for rest intervals, and writing workout logs in physical notebooks. This disconnected workflow leads to poorly matched workout intensities, inaccurate progress tracking, and high dropout rates among fitness beginners.

A unified software solution built with Python Object-Oriented Programming (OOP) principles and a relational database directly solves these inefficiencies. By encapsulating member attributes, biometric computations, and workout split hierarchies into modular classes, the application dynamically adapts exercise prescriptions to a user's unique profile while storing historical workout logs systematically in a database.

2. PROJECT DESCRIPTION

V The system automates BMI computation and health categorization, prescribes customized five-exercise routines across major training splits (Push, Pull, Legs & Core, and Full Body), and provides real-time session utilities including an active stopwatch, rest countdown timers, and a dynamic progress bar. Unlike a console program that processes text sequentially, this application features an event-driven Graphical User Interface (GUI) with interactive input forms, status indicators, clickable exercise checklists, live timers, and a data table for managing saved records.

3. OBJECTIVES

General Objective: To design and develop an object-oriented GUI application with relational database integration that automates personalized workout recommendations, tracks real-time exercise sessions, and manages member fitness logs.

Specific Objectives:

- Design and implement a graphical user interface for automated BMI calculation, dynamic routine generation, live workout timing, and progress logging.
- Apply Object-Oriented Programming principles (encapsulation, inheritance, polymorphism, abstraction) in modeling member profiles, workout splits, and calorie calculation algorithms.
- Design and integrate a relational database to store and manage user credentials, member profiles, and completed workout session histories.
- Implement core CRUD (Create, Read, Update, Delete) functionality through the GUI so users can log new workouts, view historical sessions, edit profile biometrics, and delete outdated entries.
- Ensure data validation, error handling, and basic security (e.g., login authentication and positive numeric validation for height and weight inputs) within the application.

4. SCOPE AND LIMITATIONS

Scope:

- The application will be a GUI-based desktop/web program developed using Python.
- It will include the following core modules: User Authentication, Member Profile & BMI Assessment, Smart Routine Recommender, Live Session & Rest Timers, Workout History CRUD Management, and Summary Reports.
- Data will be stored and retrieved using a SQLite / MySQL relational database.

Limitations:

- The system will not include wearable hardware synchronization (such as smartwatches), cloud deployment, or real-time multi-user network syncing unless stated otherwise.
- The generated exercise routines serve as general physical conditioning guides for healthy individuals and do not replace professional medical or physical therapy advice.

5. SYSTEM FEATURES / MODULES

Module / Feature	Description	Key OOP Class(es) Involved
Login / Authentication	Allows users to securely log in before accessing the system.	User, AuthManager
Dashboard & BMI Profiler	Displays a summary of member biometrics, computes BMI, and classifies health status.	Dashboard, MemberProfile, BMICalculator
Workout Recommender	Uses inheritance and polymorphism to generate routines based on experience level and muscle split.	WorkoutPlan, PushSplit, PullSplit, LegsSplit
Records Management	Add, view, update, and delete logged workout sessions via GUI forms and tables.	WorkoutRecord, RecordManager
Reports	Generates workout summaries and exports fitness logs to CSV/PDF formats.	ReportGenerator

6. TOOLS AND TECHNOLOGIES

Programming Language	Python
GUI Framework	Tkinter / CustomTkinter or Flask + HTML/CSS
Database	SQLite / MySQL
Database Connector Library	sqlite3 / mysql-connector-python
IDE / Editor	VS Code / PyCharm
Version Control	Git and GitHub
Other Libraries	datetime for session timestamps, csv / reportlab for PDF/CSV export

7.PROJECT TIMELINE (EXAMPLE ESTIMATED)

Phase	Activities	Target Date
Planning & Design	Requirements gathering, ERD, class diagram, UI mockups	Week 1–2
Database Setup	Create schema, tables, and sample/test data	Week 3
GUI Development	Build application windows/forms and navigation	Week 4–5
Backend / OOP Logic	Implement classes, business logic, DB integration	Week 6–7
Testing & Debugging	Unit testing, bug fixing, validation checks	Week 8
Documentation & Presentation	Finalize report, prepare demo/presentation	Week 9